

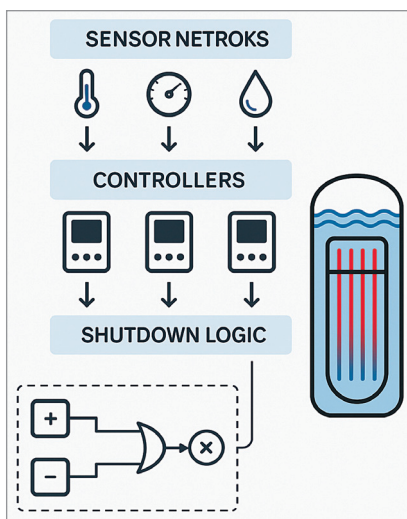


Fig. 1: A typical SMR I&C system (Source: International Atomic Energy Agency, IAEA)

being used in SMR designs. AI and ML algorithms enable predictive diagnostics and anomaly detection, allowing for proactive maintenance and minimising downtime. Advanced sensor networks monitor the real-time condition of SMRs and identify potential issues. Wireless sensor networks are being developed to reduce cabling and enhance design flexibility, particularly in space-constrained reactors. FPGA-based systems offer greater flexibility and high-performance processing capabilities for real-time data analysis and decision-making. Digital twins—virtual replicas of SMRs—are used to simulate and test reactor operations, optimise performance, and reduce the risk of errors.

The global scenario

The ongoing energy crisis, geopolitical tensions and trade uncertainties—such as the Russia-Ukraine war, the US-Russia trade conflict, and reciprocal tariffs imposed by the US—have underscored the need for secure and independent energy sources. Nuclear energy is increasingly seen as a reliable and sovereign option for delivering secure,

stable, and low-carbon power.

Several countries are progressing with SMR development. Russia has made Akademik Lomonosov—the world's first floating nuclear power plant—operational in the remote Arctic region, with a land-based unit under development by 2027. China has connected its high-temperature reactor, HTR-PM, to the grid and is expected to make it operational soon.

The United States is supporting projects such as NuScale's VOYGR, Westinghouse's AP300, and Holtec's SMR-300, with deployment timelines ranging from the late 2020s to the mid-2030s. Argentina is working towards achieving initial criticality for CAREM-32 by 2027. Canada is targeting deployment of the BWRX-300 in the 2030s for industrial decarbonisation in its remote northern mining sector.

The United Kingdom is backing Rolls-Royce's 470MWe SMR project, while South Korea is aiming to establish itself as a global manufacturing hub for SMRs. Japan and South Korea are also exploring SMRs for hydrogen production and industrial decarbonisation. The United Arab Emirates is evaluating SMRs for desalination. Space agencies such as NASA and Roscosmos are assessing SMRs as potential base power supply sources for future lunar and Martian missions.

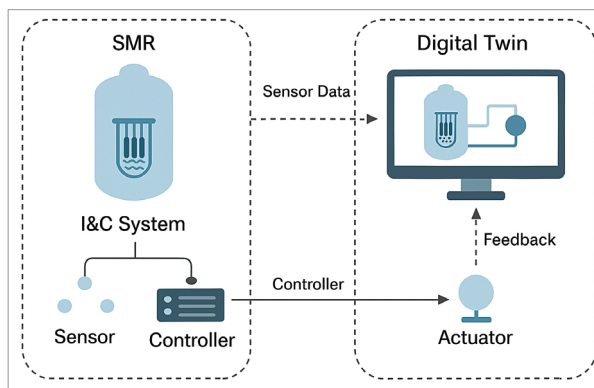


Fig. 2: Digital twins in the I&C system of SMRs

India's initiatives for SMRs

The government of India has set an ambitious target of 100GW of nuclear power by 2047, aiming to reduce carbon emissions and meet future energy demands. Recognising the transformative potential of SMRs, India has launched a series of strategic initiatives that combine indigenous innovation, regulatory reform, and international collaboration.

Strategic policy interventions and infrastructure investments are focused on developing nuclear technology domestically and fostering public-private partnerships. As part of these efforts, the Nuclear Energy Mission for Viksit Bharat has been launched to enhance national nuclear capabilities, promote private-sector participation, and accelerate the deployment of advanced nuclear technologies, such as small modular reactors (SMRs). A budget of ₹200 billion has been allocated to develop at least five indigenously designed and operational SMRs by 2033. The Atomic Energy Act and Civil Liability for Nuclear Damage Act are being amended to enable implementation of the mission and encourage private-sector involvement.

Two platforms—Bharat Small Modular Reactors (BSMRs) and Bharat Small Reactors (BSRs)—are being developed domestically.

BSMRs are 220MWe water-cooled, land-based reactors designed for grid-scale deployment. These reactors prioritise high reliability, passive safety, and seismic resistance, offering flexibility in siting. BSRs are smaller advanced reactors in the 50-100MWe range,